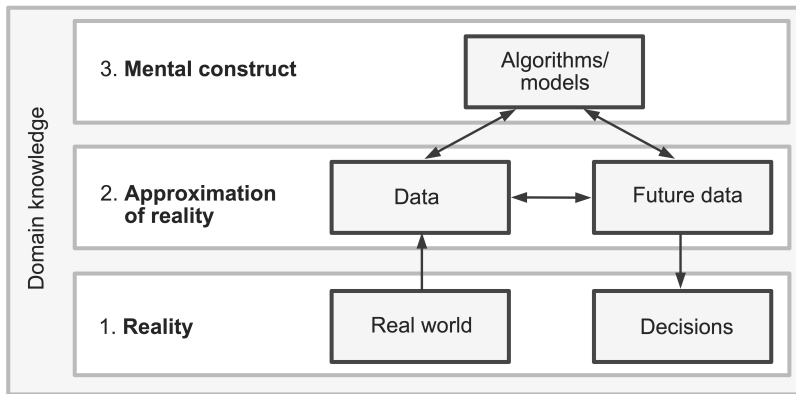


Assessing predictions

Ciaran Evans

Model assessment

Essentially, all models are wrong, but some are useful. (Box and Draper, 1987)



Today: Is our good at *predicting* the response?

Types of prediction

Regression:

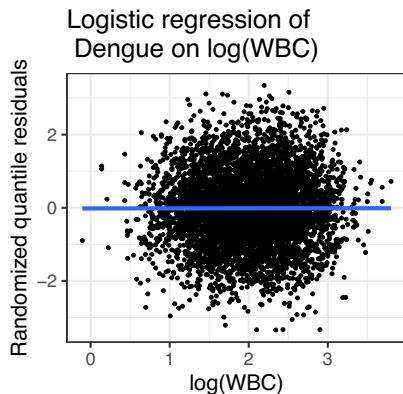
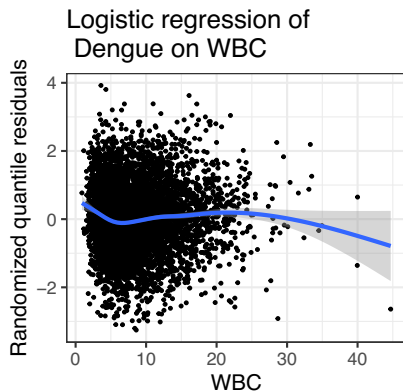
$$\begin{aligned} & y_i \in \{0, 1\} \\ \text{want } \pi_i &= P(y_i = 1) \\ \hookrightarrow \hat{\pi}_i &\in (0, 1) \quad (\text{anywhere between } 0 \text{ \& } 1) \end{aligned}$$

Classification:

$$\begin{aligned} & y_i \in \{0, 1\} \\ \text{want } \hat{y}_i & \text{ (predicted value of } y_i) \\ \hat{y}_i &\in \{0, 1\} \quad (\text{either } 0 \text{ or } 1) \end{aligned}$$

General principle: predictions from regression can be converted into classification, but not the other way around!

Assessing predicted probabilities

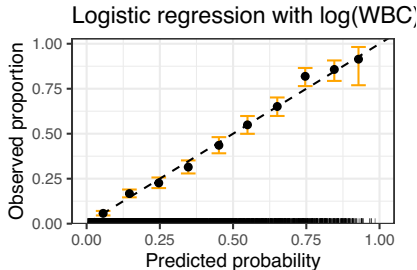
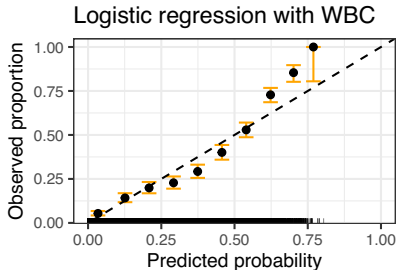
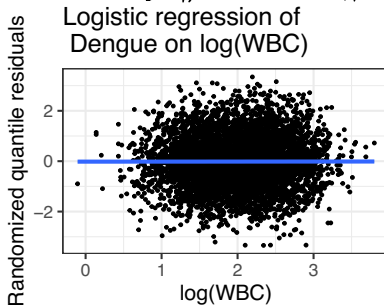
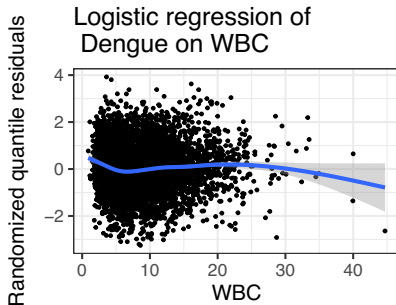


Previously : parametric bootstrap suggests WBC
does need transformation
But does it matter??

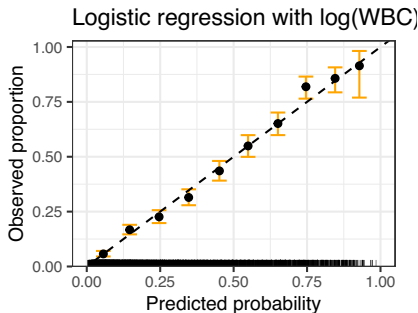
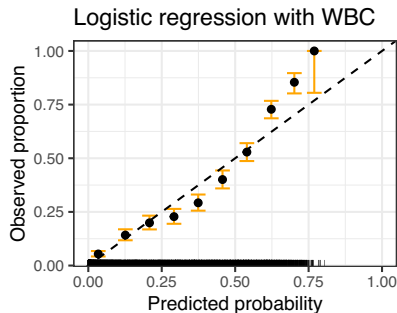
Question: Does using WBC vs. $\log(\text{WBC})$ make a difference for the quality of our predictions?

Calibration plot

- bin observations by $\hat{\pi}$
- compute observed proportion in each bin, compare to $\hat{\pi}$ from fitted model



Calibration plot

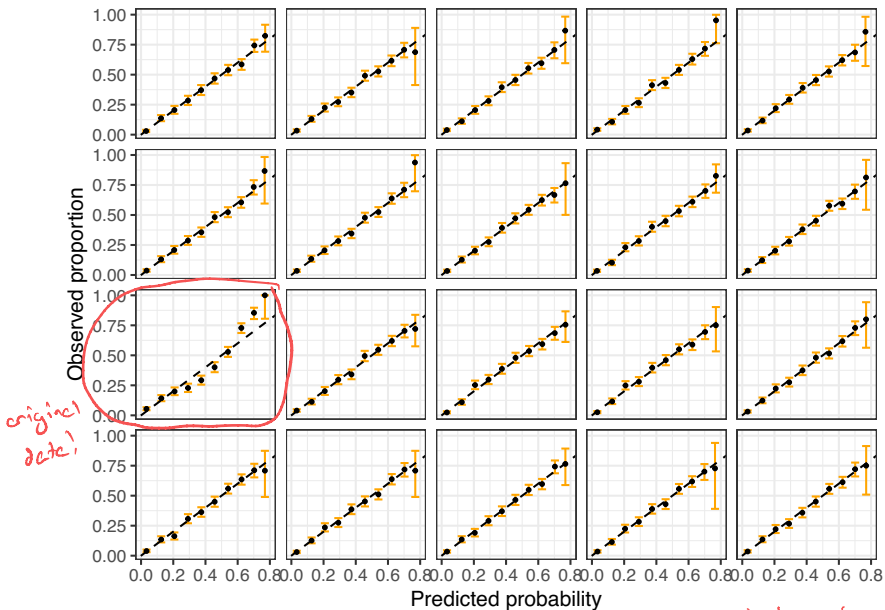


Question: How do I tell if any unusual features in the calibration plot are just due to random chance?

Parametric bootstrap:

- generate $y_1^*, \dots, y_n^*$ $y_i^* \sim \text{Bernoulli}(\hat{p}_i)$
- refit model on $(x_1, y_1^*), \dots, (x_n, y_n^*)$
- remake plots
- repeat!

Parametric bootstrap + visual inference



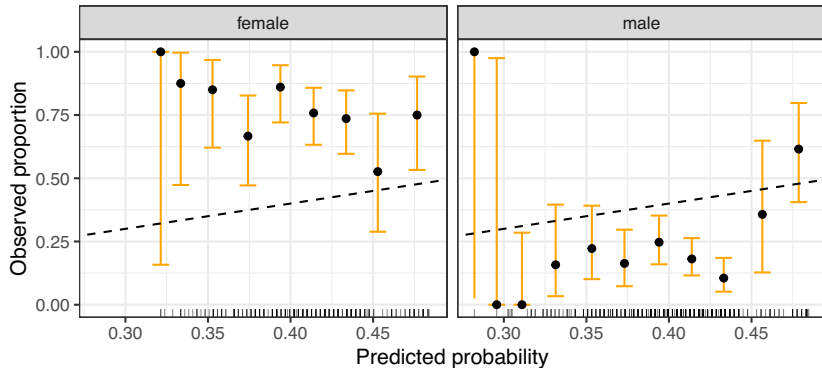
conclusion: transformation does make a difference, but maybe not by much

Class activity

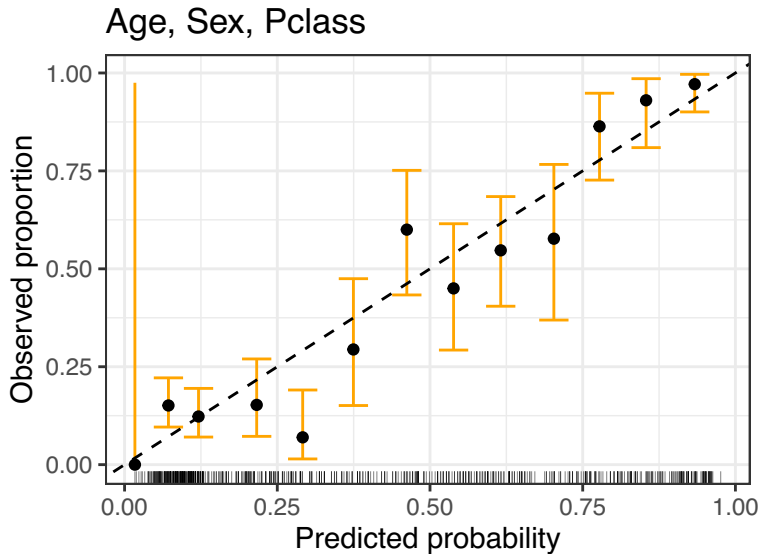
- ▶ Work on the class activity (course website) with a neighbor
- ▶ At the end of class, submit your rendered HTML on Canvas

Class activity

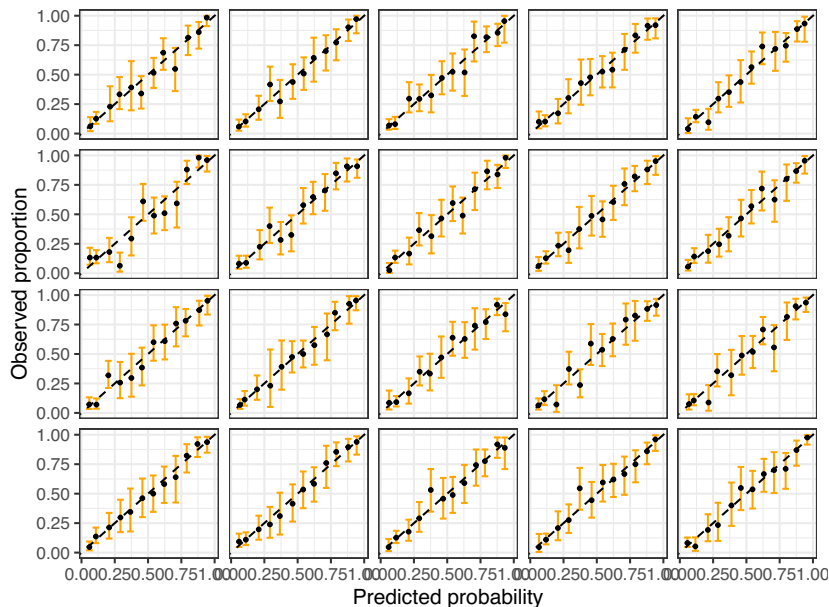
Age only, plotted by Sex



Class activity



Visual inference



Activity

Age, Sex, Pclass, plotted by Sex

